

Assignment No. 1

BT504 SPRING 2024

By M.Kashif(03064963933)

Total Marks: 10

Due date: 23-June-2024

Instructions:

- Make sure that you upload the solution file before due date. No assignment will be accepted through e-mail after the due date.

Formatting guidelines

- Use the font style "Times New Roman" and font size "12".
- Compose your document in MS-Word, any file created in any format will not be accepted and marked zero.
- Use black and blue font colours only.

Solution guidelines

- To solve this assignment, you should have good command over lectures 46 to 86.
- This is not a group assignment; it is an individual assignment so be careful and avoid copying others' work
- Give the answer according to question only and avoid irrelevant details.

Please note that your assignment will not be graded if:

- It is submitted after due date
 - The file you uploaded does not open
 - The file you uploaded is copied from someone else
 - It is in some format other than .doc
 - Cheating or copying of assignment is strictly prohibited. The cheated or copied assignment will be marked 'Zero'.
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Question1: 10 Marks

Explain in detail the (a) Working and (b) construction of Glucometer device used for measuring blood glucose levels

Glucometer: Working and Construction

A. Working of a Glucometer

A glucometer, also known as a glucose meter or blood glucose monitoring system, is a portable electronic device used to measure the concentration of glucose in the blood. The working principle of a glucometer is based on the interaction between glucose in the blood sample and specific chemicals on a test strip, which results in an electrical current that can be measured and converted into a glucose concentration reading.

1. Sample Collection

The process begins with the user obtaining a small blood sample, typically from a fingertip. A lancet device is used to prick the skin, and a tiny droplet of blood (usually about 0.3 to 1 microliter) is drawn.

2. Test Strip Insertion

Before or after obtaining the blood sample (depending on the specific device), a test strip is inserted into the glucometer. This strip contains electrodes and reagents necessary for the glucose measurement.

3. Blood Application

The blood droplet is applied to the edge of the test strip, which draws the sample into a reaction chamber through capillary action.

4. Chemical Reaction

Inside the reaction chamber, the blood sample reacts with enzymes on the test strip. Most modern glucometers use one of two enzyme systems:

- a) **Glucose Oxidase (GOx):** This enzyme catalyzes the oxidation of glucose to gluconic acid and hydrogen peroxide.
- b) **Glucose Dehydrogenase (GDH):** This enzyme catalyzes the oxidation of glucose to gluconolactone.

5. Electron Transfer

The chemical reaction produces electrons, creating an electrical current. In GOx systems, the hydrogen peroxide is electrochemically oxidized, producing electrons. In GDH systems, electrons are transferred directly from glucose to an electron mediator and then to the electrode.

6. Current Measurement

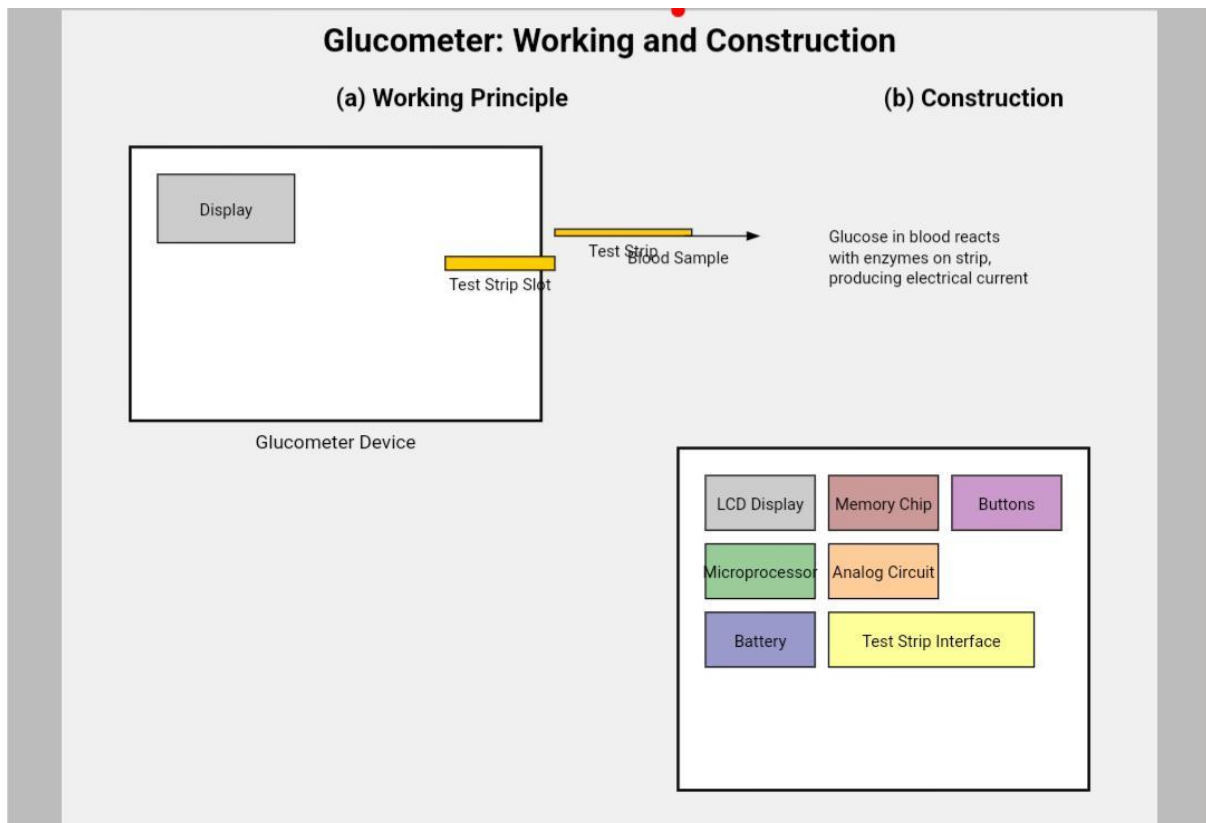
The glucometer measures the electrical current produced by the reaction. This current is proportional to the concentration of glucose in the blood sample.

7. Signal Processing and Display

The measured current is processed by the device's microprocessor, which converts it into a glucose concentration value using a calibration algorithm. This value is then displayed on the device's screen, typically in units of mg/dL (milligrams per deciliter) or mmol/L (millimoles per liter).

8. Data Storage and Management

Many modern glucometers can store multiple readings with date and time stamps. Some devices can also connect to smartphones or computers for more advanced data management and analysis.



B. Construction of a Glucometer

The construction of a glucometer involves several key components that work together to provide accurate and reliable blood glucose measurements. Here's a detailed look at the main parts:

1. Housing

The outer casing of the glucometer is typically made of durable plastic. It's designed to be compact, lightweight, and easy to hold. The housing protects the internal components and often includes ergonomic features for comfortable handling.

2. Display Screen

Most modern glucometers use a liquid crystal display (LCD) screen to show glucose readings, time, date, and other relevant information. Some advanced models may use OLED displays for better visibility.

3. Microprocessor

At the heart of the glucometer is a microprocessor that controls all functions of the device. It processes the electrical signals from the test strip, runs the calibration algorithms, manages data storage, and controls the display.

4. Test Strip Port

This is the slot where the test strip is inserted. It contains electrical contacts that connect with the electrodes on the test strip.

5. Analog-to-Digital Converter (ADC)

This component converts the analog electrical signal from the test strip into a digital signal that can be processed by the microprocessor.

6. Memory Chip

A memory chip stores glucose readings, often along with date and time stamps. The capacity can vary from a few hundred to several thousand readings.

7. Battery

Glucometers are powered by batteries, usually button cell or AAA batteries. Some models use rechargeable batteries.

8. Clock Circuit

This component keeps track of time and date for timestamping glucose readings.

9. Temperature Sensor

Many glucometers include a temperature sensor to ensure accurate readings across different environmental conditions.

10. Communication Interface

Modern glucometers often include USB ports or wireless (Bluetooth) capabilities for data transfer to computers or smartphones.

11. Control Buttons

Buttons on the device allow users to turn it on/off, navigate menus, and set various options.

12. Test Strips

While not part of the glucometer itself, test strips are crucial components of the system. They are typically made of the following parts:

- a) **Sample Application Area:** Where the blood drop is applied.
- b) **Reaction Chamber:** Contains the enzymes (GOx or GDH) and other reagents.
- c) **Electrodes:** Usually made of gold or carbon, these conduct the electrical current to the meter.
- d) **Insulating Layers:** Separate different parts of the strip.
- e) **Hydrophilic Layer:** Helps draw the blood sample into the reaction chamber.

13. Calibration System

Some glucometers require calibration to ensure accuracy. This may involve entering a code from the test strip vial or using a calibration chip.

14. Lancet Device

While not part of the glucometer itself, the lancet device is an essential accessory. It's a spring-loaded mechanism that holds a small, sharp needle (lancet) used to prick the skin for blood sampling.

15. Data Management Software

Many glucometers come with companion software for computers or smartphone apps. This software allows users to download, analyze, and share their glucose data with healthcare providers.

The construction of a glucometer involves precise engineering to ensure accurate and consistent measurements. Manufacturers must adhere to strict quality control and regulatory standards, as these devices play a crucial role in the management of diabetes. Ongoing research and development in this field continue to improve the accuracy, ease of use, and additional features of glucometers, making them increasingly valuable tools for diabetes management.

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